NATIONAL ENGINEERING COLLEGE, K.R. NAGAR, KOVILPATTI – 628 503

(An Autonomous Institution, Affiliated to Anna University – Chennai)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING &

DEPARTMENT OF INFORMATION TECHNOLOGY

23CS11E / 23IT11E - WEB FRAMEWORKS USING PYTHON

Odd Semester - 2026-2027

Preparatory Test – 01 Question bank

CO1: Demonstrate the use of Django's core programming constructs to design and implement structured solutions. (CDL1)

Part – A (Two Marks)

1. What are the common HTTP methods (GET, POST, PUT, DELETE), and how do they typically relate to CRUD operations?
2. What is Django ORM and what advantages does it offer over writing raw SQL queries?
3. Brief the purpose of makemigrations and migrate commands. When would you run each?
4. What is the purpose of Django Forms? How do they differ from simple HTML forms?
5. State the role of request.method == 'POST' in a Django view. Why is it important?
6. What does form.is_valid() will do, and why is preferred over request.POST directly?
7. What is {% csrf_token %} and why is it crucial for forms in Django?
8. Where does Django look for static files by default, and how can you tell it to look in additional directories?
9. Write a Django model snippet to define a Book model with fields: title, author, isbn.
10. Modify the following Django view to handle only POST requests:

```
def my_view(request):
    ……
    return HttpResponse("Hello")
```

11. Why are assert statements helpful during Django development?

Part – B (8 Marks)

1. Explain the core principles of RESTful architecture. Discuss the advantages of using a RESTful approach for API design. Give a python code to access and print the contents of any one open API.
2. Explain the basic flow of how a user interacts with a form on a web page and how Django processes that interaction with diagram and give an example with code?
3. Design a Django Model for a simple "Book" application. The Book model should have the following fields: title (character field, max length 200), author (character field, max length 100), publication_date (date field), and isbn (character field, unique, max length 13). Write the Python code for this model and write a python code to perform CRUD operations over shell.

4. How do you retrieve all objects, a single object, or filter objects based on conditions using the Django ORM?
5. Design a Django Model named Product for an e-commerce application. The Product model should include: name (character field, max length 255), description (text field), price (decimal field with max digits and decimal places), stock_quantity (integer field), and is_available (boolean field, default to True). Write the Python code for this model and demonstrate how to fetch all products that are available and have a stock quantity greater than 0.
6. Explain the purpose of Django Forms and how they simplify the process of handling user input. Discuss the three main components of a Django Form. Provide a simple Django forms.py code snippet for a FeedbackForm with fields for subject (CharField) and message (CharField/TextField).
7. Discuss in details of the following with neat example in Django:
    a. Model
    b. View
    c. URLs
    d. Importance of virtual environment setup
8. Write the python code for creating a form for Bus booking with name, source, destination and date of departure. Create a model as Booking to store these values. Also create views to render the form and insert values into the model.

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